

Capstone Spec: Semantic SEO & Intent Intelligence Agent

AI Fluency Track • Phase: Build (Capstone Spec) | Estimated Build: ~10 Hours

Why It Matters & Domain Context:

"The capstone is a working personal AI agent, and agents fail at the design stage more often than the build stage. A tight spec is the difference between a 10 hour build and a 30 hour swamp."

Grounding: Designed explicitly for FlyRank AI Hackathon Brief datasets (Google Search Console & GA4 data for Flewd).

1. Job to be Done & User Profile

Job to be Done (JTBD): Transform raw, multi-source search datasets (GSC site queries, GSC URL impression flags, and GA4 engagement exports) into actionable, intent-aligned SEO content optimization briefs. The agent autonomously classifies query intent, flags high-value 'striking distance' opportunities (ranks 3–15), identifies intent-engagement mismatches, and outputs ranked content recommendations.

Target User & Usage Frequency: SEO Strategists and Content Engineers working on the Flewd account. Executed weekly (and ad-hoc during sprint planning) to continuously feed the content creation pipeline with grounded growth opportunities.

2. Tools, Data Sources, and Access Plan

Data Source / Tool	Purpose in Agent Workflow	Access & Implementation Plan
GSC Site Impressions CSV	Query-level impressions, clicks, CTR, and search positions (~4,300 rows/day).	Local filesystem via Python / MCP File Server.
GSC URL Impressions CSV	Page-level SERP feature flags (Product snippets, FAQs, Merchant listings).	Local filesystem via Python / MCP File Server.
GA4 Raw Event Export	Page engagement, add-to-cart, purchase events (Join on Landing Page URL).	Local pandas script with JSON flattener for nested fields.
Embeddings / Classifier	Semantic intent classification	OpenAI / Anthropic API tool-call

Tool	(Comparison, Replacement, Risk, Use-case).	function for zero-shot intent tagging.
------	--	--

3. Draft Agent Instructions (System Prompt Core)

You are the FlyRank Intelligence Agent. Your goal is to analyze GSC and GA4 datasets for Flewd to extract strategic search opportunities.

- Step 1: Ingest GSC and GA4 datasets. Always join GSC URL data and GA4 event data using the landing page URL, NOT raw query strings (GA4 lacks organic query dimensions).
- Step 2: Deliberately isolate anonymized queries (blank query fields in ~8% site / ~36% URL data). Do not drop them; aggregate their impressions at the URL level.
- Step 3: Flatten nested JSON fields in GA4 export ('device', 'geo', 'ecommerce') to compute page-level engagement rates.
- Step 4: Classify queries into intent categories: Comparison, Replacement, Risk/Safety, or Use-case.
- Step 5: Identify striking-distance queries (Position 3.0 to 15.0 with High Impressions) and generate prioritize action briefs ranked by predicted impact.

4. Pre-Build Evaluation Cases (Five FL-03 Style Evals)

Eval ID	Input Test Case	Expected Assertions / Pass Criteria
EVAL-01	Attempt to join GSC query dataset directly with GA4 raw events on query string.	FAIL / FLAG: Agent must refuse direct query join and execute join strictly on landing page URL.
EVAL-02	GSC URL export contains 36% blank query rows.	PASS: Agent isolates blank query rows and attributes aggregate metrics directly to the target URL without crashing.
EVAL-03	Query 'magnesium taurate vs glycinate' parsed by classifier.	PASS: Categorized specifically as 'Comparison' intent, not general 'Informational'.
EVAL-04	Query ranking at Position 4.2 with 12,000 monthly impressions and 1.1% CTR.	PASS: Flagged as 'Striking Distance Opportunity' with action item 'Title/Meta Refresh'.
EVAL-05	GA4 export containing nested JSON string for mobile device ecommerce	PASS: Successfully flattens JSON without throwing parsing errors and extracts

events.	purchase revenue.
---------	-------------------

5. Risks, Guardrails, and Irreversible Action Rules

- **MUST CONFIRM (Human-in-the-loop):** Content rewrite recommendations that involve deleting or consolidating existing landing pages must require explicit human confirmation before updating site architecture.
- **NEVER DO (Absolute Block):** The agent must NEVER perform write operations on production Google Search Console or GA4 accounts. All outputs are strictly advisory recommendations saved to local briefs.
- **Data Confidentiality Guardrail:** All Flewd client data is strictly confidential; agent must not pass raw user data to unvetted external public APIs outside specified workspace models.

6. Platform Selection & Justification

Selected Platform: Scripted Python Agent with Model Context Protocol (MCP) / Claude Connectors.

Justification against Alternatives:

- Vs. Custom GPT / Claude Project: Custom GPTs lack native abilities to process nested JSON GA4 raw event exports or perform multi-dataframe pandas joins reliably across thousands of GSC rows. A scripted Python agent gives full control over data transformation.
- Vs. n8n Agent Workflow: While n8n excels at API orchestration, Python scripts with MCP provide cleaner execution for vector embeddings, HDBSCAN clustering, and localized LLM zero-shot classification within the 10-hour build limit.

7. Pass / Revise Criteria Self-Audit

Evaluation Criteria	Status	Verification Details
Scope achievable in ~10 build hours	[PASS]	Focused single job: GSC/GA4 intent modeling and opportunity spec.
Realistic access plan for tools & data	[PASS]	Local CSV/JSON access plan for GSC & GA4 Flewd datasets.
Five+ eval cases defined before building	[PASS]	5 explicit FL-03 style eval cases defined with pass/fail criteria.

Guardrails for risky / irreversible actions

[**PASS**]

Explicit 'Must Confirm' & 'Never Do' rules for data & page updates.

Platform choice justified vs. alternative

[**PASS**]

Scripted Python + MCP chosen and justified over Custom GPT & n8n.